# CS407 MCQS/QUIZ +CONCEPTUAL Q/A + SUMMARIZED NOTES + MODEL PAPER BY VULMS MASTERY

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## MCQs – Routing and Switching

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**1. The primary objective of the Routing and Switching course is to: develop deep conceptual understanding of networking principles rather than memorizing commands and facts, enabling students to apply knowledge in real-world networking scenarios.**

A. Focus on memorization of commands only
B. Develop deep conceptual understanding of networking ✓
C. Limit learning to theoretical definitions
D. Avoid practical implementation

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**2. The structure of the Routing and Switching course divides learning into three major sections, each focusing on a specific area of networking knowledge including fundamentals, routing protocols, and switching techniques.**

A. Two sections only
B. Four unrelated sections
C. Three structured networking sections ✓
D. Random topic selection

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**3. The Network Fundamentals section of this course is designed to provide students with both conceptual understanding and practical skills required to establish a strong foundation in basic networking concepts.**

A. Focus only on routers
B. Provide theoretical knowledge only
C. Combine conceptual and practical skills ✓
D. Focus only on programming

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**4. Routing protocols and concepts primarily focus on understanding the internal architecture, working components, and operational behavior of routers and routing mechanisms in networks.**

A. End-user applications
B. Router architecture and routing behavior ✓
C. Wireless configuration only
D. Database systems

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**5. Switching techniques in networking are primarily concerned with studying the structure, operation, and behavior of converged switched networks that enable efficient data communication.**

A. File storage systems
B. Converged switched network operation ✓
C. Operating systems
D. Cloud computing only

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**6. By completing this course, students are expected to gain the ability to design simple LAN networks and configure basic routing and switching devices along with IP addressing schemes.**

A. Only theoretical knowledge
B. Ability to design LAN and configure devices ✓
C. Only hardware repair
D. No practical skills



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**7. In modern communication systems, networking plays a crucial role in enabling instant global communication where individuals can share ideas, information, and resources in real time across continents.**

A. Slows communication globally
B. Enables instant global communication ✓
C. Restricts information flow
D. Works only locally

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**8. Online learning systems have significantly transformed education by removing geographical barriers and allowing students to access instructional material and courses from any location at any time.**

A. Limit access to education
B. Remove geographical barriers ✓
C. Depend on physical classrooms only
D. Restrict online collaboration

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**9. In a computer network, hosts are defined as end devices that are capable of sending and receiving data, and they may function as clients, servers, or both depending on installed software.**

A. Only servers
B. Only routers
C. End devices capable of multiple roles ✓
D. Only switches

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**10. A server in a network environment is defined as a host device that provides services such as email, web hosting, or file sharing to other devices through specialized server software.**

A. Requests data from others
B. Provides services using server software ✓
C. Only stores offline data
D. Acts as a client only

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**11. A client in a network is a computer system that uses application software such as web browsers to request and display information provided by servers on the network.**

A. Provides services
B. Requests and displays server data ✓
C. Only manages hardware
D. Acts as router

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**12. Peer-to-peer networks allow computers to function simultaneously as both clients and servers, particularly in small-scale environments such as homes and small businesses.**

A. Only client role
B. Only server role
C. Both client and server roles ✓
D. No network role

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**13. One of the major disadvantages of peer-to-peer networks is the lack of centralized administration, which makes network management and security control more difficult.**

A. High scalability
B. Centralized control
C. Lack of centralized administration ✓
D. High cost

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**14. Network infrastructure consists of three major categories including devices, media, and services, which collectively enable communication and data exchange across networks.**

A. Software only
B. Devices, media, and services ✓
C. Applications only
D. Users only

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**15. End devices in a network are those devices that directly interface with users and include systems such as computers, printers, smartphones, and IP-based communication devices.**

A. Routers only
B. User-interface devices ✓
C. Only switches
D. Only cables

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**16. Intermediary devices in a network function by connecting end devices and ensuring proper data flow, while also managing traffic and determining optimal data paths.**

A. Store files permanently
B. Connect and manage network traffic ✓
C. Only generate data
D. Replace end devices

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**17. Network media refers to the physical or wireless channels through which data is transmitted from one device to another in a communication system.**

A. Software applications
B. Data transmission channels ✓
C. Operating systems
D. User interfaces

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**18. Fiber optic cables are considered a type of network media that transmits data using light signals, allowing high-speed and long-distance communication.**

A. Electrical signals
B. Light-based transmission ✓
C. Magnetic signals
D. Sound waves

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**19. The selection of network media depends on factors such as distance, environmental conditions, data transmission speed, and installation cost.**

A. Only brand name
B. Distance, speed, cost, and environment ✓
C. Device color
D. User preference only

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**20. Physical topology diagrams are used in networking to show the actual physical layout of devices, cables, and network connections within an infrastructure.**

A. IP addressing only
B. Physical device layout ✓
C. Software design
D. Data encryption

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**21. Logical topology diagrams focus on showing how data flows within a network, including IP addressing schemes and device communication paths.**

A. Cable installation only
B. Data flow and addressing ✓
C. Physical location only
D. Power supply layout

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**22. A Local Area Network (LAN) is designed to connect devices within a limited geographical area such as a home, office building, or school campus.**

A. Global communication
B. Limited geographical area ✓
C. Satellite coverage
D. Nationwide network

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**23. A Wide Area Network (WAN) connects multiple networks over large geographical distances, often spanning cities, countries, or continents.**

A. Small office only
B. Large geographical area ✓
C. Personal device only
D. Single room

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**24. The Internet is best described as a global system of interconnected networks that communicate using standardized protocols and technologies.**



A. Single computer system
B. Network of networks ✓
C. Private LAN only
D. Offline system

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**25. Internet Service Providers (ISPs) play a crucial role in providing users with access to the Internet through various technologies such as DSL, cable, and wireless systems.**

A. Block internet access
B. Provide internet connectivity ✓
C. Store websites only
D. Manufacture devices

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**26. Cable internet delivers data using coaxial cables that also carry television signals, providing high-speed and always-on connectivity.**

A. Telephone wires
B. Coaxial cable system ✓

C. Satellite only
D. Bluetooth only

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**27. DSL technology provides internet access through telephone lines while separating voice and data channels for simultaneous usage.**

A. Fiber optic only
B. Telephone line with multiple channels ✓
C. Satellite only
D. Wireless only

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**28. Cellular internet access depends on mobile network towers and provides internet connectivity based on signal availability and device capability.**

A. Wired connection
B. Mobile network towers ✓
C. USB cables
D. Satellite dish only

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**29. Packet Tracer is a simulation tool that allows students to visualize and experiment with network topologies and device configurations in a virtual environment.**

A. Hardware router
B. Network simulation software ✓
C. Antivirus software
D. Operating system

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**30. Encoding in networking refers to the process of converting information into a format suitable for transmission over a communication medium.**

A. Deleting data
B. Converting data for transmission ✓
C. Encrypting passwords only
D. Storing files

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**31. Message encapsulation involves adding header information such as source and destination addresses to data before transmission across a network.**

A. Removing data
B. Adding addressing information ✓
C. Compressing files
D. Deleting headers

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**32. Flow control in networking ensures that data is transmitted at a rate that the receiving device can handle to prevent data loss or congestion.**

A. Increases errors
B. Controls transmission speed ✓
C. Deletes packets
D. Stops communication

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**33. Unicast communication involves sending data from one sender to one specific receiver in a network.**

A. One to many
B. One to one ✓
C. Many to many
D. Broadcast only

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**34. Multicast communication involves sending data from one sender to a selected group of receivers in a network.**

A. One to one
B. One to selected group ✓
C. One to all
D. None

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**35. Broadcast communication involves sending data from one sender to all devices within a network segment simultaneously.**

A. One to one
B. One to all devices ✓
C. One to selected
D. None

**36. TCP/IP protocol suite is an open standard model used for communication across the Internet and allows interoperability between different devices and systems.**

A. Closed system
B. Open standard protocol suite ✓
C. Hardware device
D. Software only

**37. Open standards in networking encourage innovation and prevent any single organization from monopolizing communication technologies.**

A. Restrict innovation
B. Encourage competition ✓
C. Limit development
D. Block standards

**38. Cisco IOS is the operating system used in Cisco networking devices to manage routing, switching, and network configuration tasks.**



A. Mobile OS
B. Network operating system ✓
C. Database system
D. Web browser

**39. The user EXEC mode in Cisco IOS provides limited access and does not allow configuration changes to the device.**

A. Full configuration
B. Limited monitoring access ✓
C. Root access
D. Admin control

**40. Privileged EXEC mode in Cisco IOS allows users to execute configuration commands and access advanced device management features.**

A. Read-only mode
B. Full configuration access ✓

C. Guest mode
D. Offline mode

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**41. The console port provides direct physical access to a networking device and is commonly used for initial configuration and troubleshooting.**

A. Remote access only
B. Physical local access ✓
C. Wireless access
D. Internet access

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**42. SSH provides secure remote access to networking devices by encrypting data and protecting login credentials during communication.**

A. No security
B. Encrypted secure access ✓
C. Only local access
D. No authentication

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**43. Telnet is considered less secure compared to SSH because it transmits data, including passwords, in unencrypted form.**

A. Highly secure
B. Unencrypted communication ✓
C. Encrypted only
D. Offline tool

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**44. The IOS command structure includes a command followed by optional keywords and arguments that define specific actions.**

A. Random text input
B. Structured command format ✓
C. Binary input only
D. GUI commands only

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**45. Context-sensitive help in IOS is accessed using a question mark and provides available commands based on the current mode.**

A. Deletes commands
B. Shows available commands ✓
C. Installs software
D. Restarts device

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**46. Command syntax errors in IOS can occur due to incomplete commands, ambiguous commands, or incorrect command usage.**

A. Hardware failure
B. Syntax-related errors ✓
C. Network failure
D. Power failure

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**47. Hot keys in IOS such as Ctrl+C and Ctrl+Z help users efficiently navigate and control command execution.**

A. Delete system files
B. Improve navigation efficiency ✓
C. Shutdown device
D. Format disk

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**48. The TAB key in IOS helps complete partially typed commands automatically if the command is unique.**



A. Deletes input
B. Auto-completes commands ✓
C. Saves files
D. Restarts system

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**49. The enable command in IOS is used to switch from user EXEC mode to privileged EXEC mode.**

A. Disable system
B. Enter privileged mode ✓
C. Exit device
D. Restart router

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**50. IOS global configuration mode allows changes that affect the entire device configuration and overall network behavior.**

A. Single interface only
B. Device-wide configuration ✓
C. Only monitoring
D. No changes allowed

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**1. One of the most frequently used Cisco IOS commands such as "show version" is primarily used on routers and switches to display detailed system information including IOS version, hardware specifications, and device status for troubleshooting and identification purposes.**

A. Deletes IOS file
B. Displays system and IOS information ✓
C. Configures IP address
D. Resets device

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**2. The "show version" command is especially useful in network environments where administrators need to quickly gather summarized information about a remotely connected Cisco device without physically accessing it.**

A. Used only locally
B. Used for password reset
C. Used for remote device information retrieval ✓
D. Used for cable configuration

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**3. Cisco IOS navigation activities in Packet Tracer are designed to help learners understand user EXEC mode, privileged EXEC mode, and configuration modes through practical simulation-based interaction.**

A. Hardware repair practice
B. IOS mode understanding and navigation ✓
C. Only IP addressing
D. Only cable selection

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**4. Cisco devices such as routers and switches share similar operating systems and command structures, which allows network administrators to configure them using consistent IOS-based commands across multiple platforms.**

A. Completely different systems
B. Similar IOS structure and commands ✓

C. No shared features
D. Hardware-only configuration

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**5. A Cisco switch can operate with minimal configuration and still forward data between connected devices because its default behavior allows basic Layer 2 switching functionality without requiring initial setup.**

A. Requires full configuration
B. Works without configuration ✓
C. Requires routing table
D. Needs internet access

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**6. Assigning a hostname to a Cisco device is an essential initial configuration step because it allows identification in CLI prompts, network documentation, and authentication processes between devices.**

A. Only for encryption
B. For device identification ✓
C. For hardware upgrade
D. For IP routing only

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**7. Hostnames in Cisco IOS must follow strict naming conventions such as starting with a letter, containing no spaces, and being less than 64 characters to ensure proper device recognition and system compatibility.**

A. Any random format allowed
B. Must follow naming rules ✓
C. Only numbers allowed
D. Must be IP address

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**8. The enable secret command is preferred over enable password because it provides encrypted password storage, enhancing security in privileged EXEC mode access.**

A. Less secure method
B. Encrypted and secure method ✓
C. No password needed
D. Only for switches

**9. Securing the console port of a Cisco device is important because it prevents unauthorized physical access that could allow direct configuration or control of the network device.**

A. No security needed
B. Prevents unauthorized access ✓
C. Only for internet access
D. Only for remote login

**10. VTY lines on Cisco devices are used to allow remote access through Telnet sessions, enabling administrators to configure devices over a network connection.**

A. Local console access
B. Remote Telnet access ✓
C. Hardware reset
D. Firewall configuration



**11. Packet Tracer initial switch configuration activities typically involve setting passwords, creating login banners, and securing CLI access to prevent unauthorized usage of network devices.**

A. Only IP routing
B. Device security configuration ✓
C. Hardware replacement
D. Wireless setup only

**12. Implementing basic connectivity in a network involves configuring IP addresses on switches and PCs and verifying communication using commands such as ping and show.**

A. Only cable connection
B. IP configuration and verification ✓
C. Antivirus installation
D. Operating system setup

**13. Physical layer connectivity in a network can be established using wired connections like copper or fiber cables or wireless connections depending on network design requirements.**

A. Only wireless
B. Wired and wireless media ✓
C. Only Bluetooth
D. Only software links

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**14. A Network Interface Card (NIC) is responsible for connecting a host device to a network using either wired Ethernet or wireless WLAN technology.**

A. Stores files
B. Provides network connectivity ✓
C. Routes packets
D. Encrypts data only

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**15. Copper, fiber optic, and wireless are the three primary media types used in physical layer communication for transmitting data signals in networks.**

A. Only copper
B. Three main media types ✓
C. Only satellite
D. Only Bluetooth



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**16. Bandwidth in networking refers to the maximum capacity of a communication medium to transmit data over a given period of time, typically measured in bits per second.**

A. Physical cable length
B. Data transmission capacity ✓
C. Number of users
D. Router speed only

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**17. Throughput refers to the actual rate at which data is successfully transmitted across a network medium and is usually lower than theoretical bandwidth due to various network factors.**

A. Maximum possible speed
B. Actual data transfer rate ✓
C. Cable type
D. Router model

**18. Latency in a network refers to the time delay experienced while data travels from the source to the destination across one or more network devices.**

A. Data storage size
B. Transmission delay ✓
C. IP address type
D. Packet format

**19. Unshielded Twisted Pair (UTP) cables are widely used in LAN environments due to their low cost and ability to reduce interference through wire twisting.**

A. Expensive and rare
B. Common LAN cabling ✓
C. Wireless medium
D. Satellite medium

**20. Shielded Twisted Pair (STP) cables provide better protection against electromagnetic interference but are more expensive and harder to install than UTP cables.**



A. Cheaper than UTP
B. Better EMI protection ✓
C. Wireless cable
D. No shielding required

**21. Coaxial cable consists of a central conductor surrounded by insulation and shielding layers that reduce electromagnetic interference and improve signal quality.**

A. Single wire only
B. Multi-layer shielding structure ✓
C. Wireless medium
D. Fiber-based cable

**22. Fiber optic cables transmit data using light signals, enabling high-speed, long-distance communication with minimal signal loss and immunity to electromagnetic interference.**

A. Electrical signals
B. Light-based transmission ✓
C. Radio waves
D. Copper signals only

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**23. Enterprise networks use fiber optic cables mainly for backbone connections between major network devices to support high-speed data transfer.**

A. Only home networks
B. Backbone enterprise use ✓
C. Only wireless
D. Only mobile networks

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**24. The Data Link Layer is responsible for packaging network layer packets into frames and controlling how data is placed on and retrieved from the physical media.**

A. IP addressing
B. Frame creation and media access ✓
C. Web browsing
D. Email services

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**25. The Logical Link Control (LLC) sublayer identifies which network layer protocol is being used to ensure multiple Layer 3 protocols can share the same network interface.**

A. Physical cabling
B. Protocol identification ✓
C. Router hardware
D. IP addressing only

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**26. The Media Access Control (MAC) sublayer handles hardware-level addressing and controls how devices access the physical transmission medium.**

A. Software configuration
B. Hardware media access ✓
C. DNS resolution
D. Routing decisions

**27. Frames in the data link layer contain source and destination addresses, allowing proper delivery of data across network segments.**

A. Only data
B. Addressing and control info ✓
C. Only IP address
D. Only encryption data

**28. The Frame Check Sequence (FCS) field is used for error detection by comparing CRC values at sender and receiver ends.**

A. Data encryption
B. Error detection ✓
C. IP routing
D. Packet switching

**29. The network layer is responsible for addressing, encapsulation, routing, and de-encapsulation of packets across different networks.**

A. Only switching
B. End-to-end packet delivery ✓
C. Only physical transmission
D. Only encryption



**30. IPv4 packets consist of a header containing control information and a payload containing transport layer data.**

A. Only headers
B. Header and payload ✓
C. Only data
D. Only IP address

**31. The Time-to-Live (TTL) field in IPv4 prevents packets from circulating indefinitely by reducing its value at each router hop.**

A. Stores IP address
B. Limits packet lifetime ✓
C. Encrypts data
D. Defines protocol type

**32. Private IPv4 addresses are used within internal networks and are not routable on the public Internet.**

A. Internet routable
B. Internal network use ✓
C. Global access
D. Broadcast only

**33. The loopback address (127.0.0.1) is used for testing local host communication without involving external network devices.**

A. External routing
B. Local system testing ✓
C. Internet browsing
D. DNS service

**34. NAT is commonly used to convert private IP addresses into public IP addresses for Internet communication.**

A. Encrypts data
B. Address translation ✓
C. Packet routing
D. DNS resolution



**35. Subnetting is the process of dividing a large network into smaller logical networks to improve performance and reduce broadcast traffic.**

A. Increasing traffic
B. Network division ✓
C. Hardware upgrade
D. Encryption method

**36. A subnet mask determines which portion of an IP address represents the network and which represents the host.**

A. Device type
B. Network vs host portion ✓
C. Internet speed
D. Cable type

**37. Routers are required for communication between different subnets because they operate at the network layer and direct traffic between networks.**

A. Switch-level device
B. Inter-network communication ✓
C. Only local traffic
D. Wireless device only

**38. Unicast communication involves sending data from one source device directly to a single destination device.**

A. One to many
B. One to one ✓
C. One to all
D. Group communication

**39. Broadcast communication sends data to all devices within a network broadcast domain simultaneously.**



A. Single device only
B. All devices ✓
C. Random device
D. External network only

**40. Multicast communication sends data to a selected group of devices that are members of a specific multicast group.**

A. All devices
B. Selected group ✓
C. Single device
D. No device

**41. Cisco IOS is a network operating system used to manage routing, switching, security, and device configuration in Cisco hardware.**

A. Desktop OS
B. Network OS ✓

C. Mobile OS
D. Database OS

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**42. RAM in a router stores running configuration, routing tables, and temporary packet buffers during operation.**

A. Permanent storage
B. Temporary operational memory ✓
C. Offline storage
D. External storage

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**43. ROM in a router contains boot instructions and basic diagnostic software required during system startup.**

A. User files
B. Boot and diagnostics ✓
C. Routing tables
D. Configuration files

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**44. NVRAM stores the startup configuration file which remains intact even when the router is powered off.**

A. Temporary data
B. Permanent configuration ✓
C. Routing data only
D. Packet data

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**45. Flash memory in routers is used to store IOS software and system files, which can be upgraded when needed.**

A. Temporary memory
B. IOS storage ✓
C. Packet buffer
D. Cache memory

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**46. The console port provides direct physical access to a router or switch for initial configuration and troubleshooting.**

A. Remote access
B. Direct local access ✓
C. Internet access
D. Wireless access

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**47. SSH is preferred over Telnet because it encrypts communication and provides secure remote device access.**

A. Less secure
B. Encrypted secure access ✓
C. No authentication
D. Offline tool

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**1. A router de-encapsulates the Layer 2 frame before processing the Layer 3 packet.**
A. Encapsulates packet
B. Removes Layer 2 frame ✓
C. Changes IP address
D. Deletes routing table

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**2. Layer 3 IP addresses remain unchanged while a packet travels across multiple routers.**
A. Change at every hop
B. Remain unchanged ✓
C. Are deleted
D. Are encrypted

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**3. Layer 2 MAC addresses change at every hop in a routed network.**
A. Remain same end-to-end
B. Change at every hop ✓
C. Are removed permanently
D. Only change at destination

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**4. A router forwards packets directly if the destination is on a directly connected network.**
A. Sends to ISP
B. Uses another router
C. Forwards directly ✓
D. Drops packet always

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**5. If no route is found, the router sends packets to a Gateway of Last Resort.**
A. Deletes packet
B. Broadcasts packet
C. Uses default route ✓
D. Stores packet

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**6. If no default route exists and no match is found, the packet is discarded.**
A. Forwarded anyway
B. Sent to all routers
C. Discarded ✓
D. Converted to broadcast

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**7. RIP uses hop count as its routing metric.**
A. Bandwidth
B. Delay
C. Hop count ✓
D. Load

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**8. OSPF calculates cost based on cumulative bandwidth.**
A. Hop count
B. Bandwidth cost ✓
C. Packet size
D. Latency only

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**9. EIGRP uses bandwidth, delay, load, and reliability as metrics.**
A. Only hop count
B. Multiple metrics ✓
C. Only bandwidth
D. Only delay

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**10. Equal cost load balancing allows multiple paths with same metric.**
A. One path only
B. Multiple equal paths ✓
C. No routing
D. Static routing only

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**11. Static routes automatically adjust when network topology changes.**
A. Yes always
B. No ✓

C. Only in OSPF
D. Only in RIP

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**12. Dynamic routing protocols automatically update routing tables.**
A. Manual only
B. Automatic ✓
C. Never update
D. Only static routes

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**13. The routing table contains directly connected and remote networks.**
A. Only remote
B. Only local
C. Both ✓
D. Only static

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**14. The "C" code in routing table represents directly connected routes.**
A. Static route
B. Connected route ✓
C. Dynamic route
D. Default route

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**15. The "S" code in routing table represents static routes.**
A. Connected route
B. Static route ✓
C. Dynamic route
D. Local route

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**16. ICMP is used for error reporting in IP networks.**
A. Routing protocol
B. Error messaging ✓
C. Switching method
D. Encryption tool

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**17. Ping uses ICMP echo request and reply messages.**
A. TCP packets
B. ICMP packets ✓
C. HTTP packets
D. UDP packets

**18. Traceroute shows the path packets take across a network.**
A. Only destination IP
B. Path of hops ✓
C. MAC addresses only
D. DNS records

**19. A floating static route acts as a backup route with higher administrative distance.**
A. Primary route
B. Backup route ✓
C. Default route
D. Dynamic route

**20. A default route is used when no specific route matches the destination.**
A. Always first choice
B. Last resort route ✓
C. Only LAN route
D. Only static route

**1. Each router learns about its directly connected networks by detecting that an interface is in the up state.**
A. Through routing table exchange
B. By static configuration only
C. By detecting interface up state ✓
D. From ISP updates

**2. In link-state routing, routers discover neighbors using Hello packets.**
A. ICMP packets
B. ARP requests
C. Hello packets ✓
D. TCP segments

**3. A neighbor relationship in link-state routing is formed when routers establish adjacency.**
A. Routes are deleted
B. Interfaces are shut down
C. Adjacency is formed ✓
D. IP is changed

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**4. Each router builds a Link-State Packet (LSP) containing link information.**
A. Routing table summary
B. Interface configuration file
C. Link-State Packet ✓
D. ARP table

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**5. LSP contains information such as neighbor ID, link type, and bandwidth.**
A. Only IP address
B. Only MAC address
C. Neighbor ID, link type, bandwidth ✓
D. Only subnet mask

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**6. LSPs are flooded to all neighbors in the routing area.**
A. Sent only to ISP
B. Stored locally only
C. Flooded to all neighbors ✓
D. Sent via FTP



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**7. Each router stores received LSPs in a link-state database.**
A. Cache memory
B. Routing table only
C. Link-state database ✓
D. ROM

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**8. The SPF algorithm is used to calculate the best path in link-state routing.**
A. RIP algorithm
B. Dijkstra's SPF algorithm ✓
C. Bellman-Ford only
D. Flooding algorithm

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**9. LSPs are not sent periodically in link-state routing protocols.**
A. Sent every 30 seconds
B. Sent every 90 seconds
C. Only when topology changes ✓
D. Sent every minute

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**10. LSP flooding occurs when topology changes or router restarts.**
A. Only during reboot
B. Only during failures
C. During restart or topology change ✓
D. Every second

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**11. Hello packets are used to maintain neighbor relationships in link-state routing.**
A. Delete routing tables
B. Maintain adjacency ✓
C. Encrypt data
D. Assign IP addresses

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**12. If Hello packets are not received, the neighbor is considered unreachable.**
A. Router is upgraded
B. Neighbor is unreachable ✓
C. IP is changed
D. Routing stops permanently

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**13. Each router independently computes the shortest path using SPF algorithm.**
A. Central server computes path
B. Router independently computes SPF ✓
C. ISP computes route
D. Switch computes route



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**14. The link-state database contains all received LSPs from neighbors.**
A. Only default routes
B. Only static routes
C. All received LSPs ✓
D. Only MAC addresses

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**15. Link-state routing protocols build a complete map of the network topology.**
A. Partial view only
B. No topology view
C. Complete topology map ✓
D. Random routing table

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# Conceptual Questions & Answers (1–50)

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### 1. What happens inside a router when it receives a packet from another network?

A router first removes the Layer 2 frame header and trailer to extract the Layer 3 packet. Then it examines the destination IP address, consults its routing table to find the best path, and finally re-encapsulates the packet into a new Layer 2 frame before forwarding it to the next hop.

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### 2. Why do Layer 2 addresses change at every hop but Layer 3 addresses remain the same?

Layer 3 IP addresses identify the source and final destination and remain unchanged end-to-end. However, Layer 2 MAC addresses are only valid within a local network segment, so they change at every router hop as the frame is rebuilt.

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### 3. What are the three main steps performed by a router when forwarding a packet?

(1) De-encapsulation of Layer 2 frame,
(2) Routing table lookup using destination IP,
(3) Re-encapsulation into a new Layer 2 frame for forwarding.



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### 4. How does a router determine the best path to a destination network?

It searches its routing table for matching entries and selects the route with the lowest metric. Metrics may include hop count, bandwidth, delay, or cost depending on the routing protocol.

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### 5. What happens if a router finds no route to a destination?

If no route is found, the router checks for a default route (Gateway of Last Resort). If none exists, it discards the packet and sends an ICMP unreachable message to the source.

---

### 6. What is a directly connected network in a routing table?

It is a network that is directly attached to one of the router's active interfaces. The router automatically adds it when the interface is configured and enabled.

---

### 7. What is a remote network in routing?

A remote network is a network not directly connected to the router. It can only be reached through another router.

---

### 8. What is meant by routing metric?

A metric is a numerical value used by routing protocols to determine the best path to a destination network. Lower metrics indicate better paths.

---

### 9. How does RIP determine the best path?

RIP uses hop count as its metric. The route with the least number of hops is selected as the best path.

---

### 10. What metric does OSPF use?

OSPF uses cost, which is typically based on bandwidth of the links along the path.

---

### 11. What metrics does EIGRP use?

EIGRP uses a combination of bandwidth, delay, load, and reliability.

---

### 12. What is equal-cost load balancing?

It occurs when multiple routes to the same destination have equal metrics. The router uses all paths equally to forward traffic.

---

### 13. What is a routing table?

A routing table is a data structure stored in RAM that contains information about directly connected and remote networks and the best paths to reach them.

### 14. What does the "C" code in a routing table indicate?

It indicates a directly connected network.

### 15. What does the "L" code represent in a routing table?

It represents the local interface IP address of the router.

### 16. What does the "S" code indicate in routing?

It represents a static route manually configured by the administrator.

### 17. What is a static route?

A static route is manually configured by a network administrator to define a fixed path to a destination network.

### 18. What is a default static route?

It is a route that matches all unknown destinations (0.0.0.0/0) and forwards packets to a gateway of last resort.

### 19. What is the advantage of static routing?

It is simple, secure, uses no CPU for route calculation, and consumes no bandwidth for updates.

### 20. What is the disadvantage of static routing?

It does not automatically adapt to network changes and must be manually updated.

### 21. What is dynamic routing?

Dynamic routing uses routing protocols to automatically discover networks, update routing tables, and adapt to topology changes.

---

## 22. What is network convergence?

Convergence is the state when all routers have complete and consistent routing information about the network.

---

## 23. Why is fast convergence important?

Because until convergence is achieved, routers may forward incorrect or incomplete routing information.

---

## 24. What is split horizon in routing?

It is a loop prevention technique that prevents a router from sending route information back out the interface from which it was learned.

---

## 25. What is ICMP used for?

ICMP is used for error reporting and diagnostic functions such as ping, unreachable messages, and time exceeded notifications.

---

## 26. What does a ping test do?

Ping tests connectivity between devices using ICMP echo request and echo reply messages.

---

## 27. What does a successful ping indicate?

It indicates that the destination host and network path are reachable and operational.

---

## 28. What is traceroute used for?

Traceroute shows the path (hops) packets take to reach a destination and identifies where delays or failures occur.

---

### 29. How does traceroute work?

It sends packets with increasing TTL values; each router decrements TTL and returns ICMP time exceeded messages.

---

### 30. What is TTL in networking?

Time To Live (TTL) limits how long a packet can exist in the network before being discarded.

---

### 31. What is hop limit in IPv6?

It is the equivalent of TTL in IPv6, controlling how many routers a packet can traverse.

---

### 32. What is a distance vector routing protocol?

It is a routing protocol where routers share distance (metric) and direction (next hop) with neighbors.

---

### 33. What is a link-state routing protocol?

It builds a complete map of the network using link-state advertisements and computes best paths using SPF algorithm.

---

### 34. What is OSPF based on?

OSPF is based on Dijkstra's Shortest Path First (SPF) algorithm.

---

### 35. What is the main difference between RIP and OSPF?

RIP uses hop count and periodic updates, while OSPF uses cost and only sends updates when topology changes.

---

### 36. What is a routing protocol?

It is a set of rules and processes that routers use to exchange routing information and determine best paths.

---

### 37. What is an autonomous system?

It is a collection of routers under a single administrative control.

---

### 38. What is IGP?

Interior Gateway Protocol is used for routing within an autonomous system.

---

### 39. What is EGP?

Exterior Gateway Protocol is used for routing between autonomous systems (e.g., BGP).

---

### 40. What is route summarization?

It is the process of combining multiple networks into a single route advertisement to reduce routing table size.

---

### 41. What is CIDR?

Classless Inter-Domain Routing allows flexible IP address allocation and supports route summarization.

---

### 42. What is a floating static route?

It is a backup static route with a higher administrative distance than the primary route.

---

### 43. What is administrative distance?

It is a value that rates the trustworthiness of a routing source; lower values are preferred.

---

### 44. What is a next-hop static route?

It specifies only the next router IP address to reach a destination.

---

### 45. What is a fully specified static route?

It includes both next-hop IP address and exit interface for clarity and efficiency.

---

### 46. What is a directly connected static route?

It specifies only the exit interface instead of next-hop IP.

---

### 47. What is the purpose of link-state updates?

They inform routers about network topology changes so all routers maintain an updated map.

---

### 48. What is SPF algorithm used for?

It calculates the shortest and most efficient path to each network in link-state routing.

---

### 49. Why is RIP considered inefficient for large networks?

Because it sends periodic full routing updates and uses hop count only, limiting scalability.

---

### 50. What happens when a network topology changes in dynamic routing?

Routers detect the change, update routing tables, and propagate new routing information automatically to maintain convergence.

---

# Summarized Networking Notes (Complete Overview)

Routers operate at the Network Layer and are responsible for forwarding packets between different networks. When a packet arrives, the router removes the Layer 2 frame, checks the destination IP address, searches its routing table for the best path, and then encapsulates the packet into a new Layer 2 frame before forwarding it. IP addresses remain unchanged from source to destination, but MAC addresses change at every hop because each router re-encapsulates the packet for the next link.

Routing decisions depend on the routing table, which contains directly connected routes, remote routes, and default routes. Directly connected networks are automatically added when interfaces are configured and activated. Remote networks are learned either through static configuration or dynamic routing protocols. If no matching route is found, the router uses a default route (Gateway of Last Resort) or drops the packet and sends an ICMP unreachable message.

Routers use metrics to choose the best path. Different routing protocols use different metrics: RIP uses hop count, OSPF uses cost based on bandwidth, and EIGRP uses a combination of bandwidth, delay, reliability, and load. When multiple equal-cost paths exist, routers use load balancing to forward traffic across all available paths.

Routing tables store different route types identified by codes such as C (connected), L (local), S (static), D (EIGRP), and O (OSPF). Static routes are manually configured and do not change automatically. They include standard static routes, default static routes, summary routes, and floating static routes used for backup with higher administrative distance. Dynamic routing protocols automatically learn and update routes as network topology changes.



Dynamic routing enables routers to exchange routing information and automatically adapt to changes. It supports network discovery, routing table updates, and best path selection. Networks are considered converged when all routers have consistent routing information. Distance vector protocols like RIP share periodic updates with neighbors, while link-state protocols like OSPF build a complete topology map and use Dijkstra's SPF algorithm to calculate the shortest path. Link-state protocols converge faster and scale better than distance vector protocols.

Routing protocols are categorized as IGP (used within an autonomous system) such as RIP, EIGRP, OSPF, and IS-IS, and EGP (used between autonomous systems), where BGP is the only widely used protocol on the internet.

ICMP is used for diagnostic and error reporting functions in networks. It provides messages such as destination unreachable, time exceeded, and echo request/reply. The ping command uses ICMP to test connectivity between hosts and measure response time. Traceroute uses ICMP and TTL/hop limit values to identify each router hop along the path and detect where delays or failures occur.

Advanced routing concepts include route summarization (CIDR), which reduces routing table size by combining multiple networks into one route, improving efficiency and reducing overhead. Floating static routes provide backup paths that are only used when the primary route fails. Default routes act as catch-all routes when no specific path exists.

Link-state routing protocols such as OSPF operate by discovering neighbors using Hello packets, exchanging link-state packets (LSPs), flooding updates across the network, and building a complete topology database. Each router independently runs the SPF algorithm to calculate the best path to every destination.

Overall, routing ensures efficient and reliable delivery of data across interconnected networks using a combination of static routes, dynamic routing protocols, path selection algorithms, and diagnostic tools, ensuring network stability, scalability, and performance.

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# MODEL PPER (Tricky & Conceptual Networking Exam)

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## Section A: MCQs



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**1. A router receives a packet and forwards it to another network. Which statement is ALWAYS true?**
A. IP address changes at every hop
B. MAC address remains same end-to-end
C. Layer 2 header changes at each hop ✓
D. Layer 4 header changes at router

---

**2. Why does a router re-encapsulate a packet at every hop?**
A. To change IP address
B. To match new Layer 2 network ✓
C. To encrypt data
D. To reduce TTL

---

**3. If a routing table has no entry for a destination and no default route exists, what happens?**
A. Packet is broadcast
B. Packet is forwarded anyway
C. Packet is dropped and ICMP sent ✓
D. Packet is stored

---

**4. Which best describes a routing metric?**
A. Physical distance between routers
B. Value used to determine best path ✓
C. IP address length
D. MAC address priority

---

**5. Two equal-cost routes exist. What will router do?**
A. Choose fastest manually
B. Drop one route
C. Load balance ✓
D. Use only first route learned

---

**6. RIP prefers a path with:**
A. Lowest bandwidth
B. Fewest hops ✓
C. Lowest delay
D. Highest reliability

---

**7. OSPF chooses best path based on:**
A. Hop count
B. Cost derived from bandwidth ✓
C. Random selection
D. MAC address order

---

**8. Why is hop count considered a weak metric?**
A. It considers bandwidth
B. It ignores link speed ✓
C. It changes dynamically
D. It uses IP addresses

---

**9. What is TRUE about Layer 3 IP addresses during routing?**
A. They change at every hop
B. They remain constant end-to-end ✓
C. They are removed
D. They are replaced with MAC

---

**10. Which entry type is automatically created when interface is configured?**
A. Static route
B. Dynamic route

C. Connected route ✓
D. Default route

---

**11. What does "Gateway of Last Resort" mean?**
A. First router
B. Backup switch
C. Default route ✓
D. Loopback interface

---

**12. Which protocol is NOT distance vector?**
A. RIP
B. EIGRP
C. OSPF ✓
D. IGRP

---

**13. Distance vector routing is mainly based on:**
A. Full topology knowledge
B. Neighbor information only ✓
C. Internet map
D. DNS records

---

**14. Link-state routing builds network using:**
A. Hop counting
B. LSP flooding ✓
C. Broadcast only
D. MAC tables

---

**15. What triggers link-state updates?**
A. Every 30 seconds
B. Only topology change ✓
C. Every packet
D. Manual request only

---

**16. Convergence means:**
A. Router reset
B. All routers agree on topology ✓
C. Packet forwarding stops
D. Routing table clears

**17. Slow convergence causes:**
A. Faster routing
B. Temporary routing loops ✓
C. Lower CPU usage
D. Better security

**18. What ICMP is mainly used for?**
A. Data encryption
B. Error reporting & diagnostics ✓
C. Routing decisions
D. Switching frames

**19. Ping failure always means:**
A. Destination is down
B. Network issue or filtering ✓
C. IP is wrong always
D. Router is off

**20. Traceroute uses which field to discover hops?**
A. MAC address
B. TTL / Hop Limit ✓
C. IP header checksum
D. Port number

**21. Each router hop in traceroute:**
A. Increases TTL
B. Decreases TTL ✓
C. Resets TTL
D. Ignores TTL

**22. Routing table contains:**
A. Only IP addresses
B. Only MAC addresses
C. Networks + next hop info ✓
D. DNS entries

**23. Static routing is preferred when:**
A. Network changes often
B. Network is small and stable ✓
C. Internet routing
D. Wireless networks

---

**24. Dynamic routing advantage is:**
A. Manual control
B. Automatic adaptation ✓
C. No CPU usage
D. No updates

---

**25. Floating static route is activated when:**
A. Primary route works
B. Primary route fails ✓
C. Router boots
D. Ping is successful

---

**26. Administrative distance defines:**
A. Speed
B. Trust level of route source ✓
C. Packet size
D. Bandwidth

---

**27. Which route is MOST preferred?**
A. Higher AD
B. Lower AD ✓
C. Random AD
D. Equal AD always

---

**28. Route summarization helps in:**
A. Increasing routes
B. Reducing routing table size ✓
C. Increasing hops
D. Slowing convergence

---

**29. BGP is mainly used for:**
A. LAN routing
B. Internet routing between AS ✓

C. Switch configuration
D. Local routing only

---

**30. Split horizon is used to prevent:**
A. IP conflict
B. Routing loops ✓
C. Packet loss
D. Encryption

---

**31. RIP sends updates:**
A. Only on change
B. Every 30 seconds ✓
C. Every hour
D. Manually only

---

**32. OSPF is preferred over RIP because:**
A. Uses hop count
B. Faster convergence ✓
C. Simpler design
D. No routing table

---

*33. If traceroute shows “ * ”, it means:*
A. Fast response
B. Packet lost or blocked ✓
C. IP resolved
D. Route successful



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# Section B: Short Questions

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### Q1: Why do routers need routing tables instead of static memory paths?

Routers operate in dynamic environments where networks change frequently. Routing tables allow flexible, updated path selection based on real-time topology changes, unlike fixed memory paths.

---

### Q2: Why is ICMP not considered a reliable transport protocol?

ICMP only reports errors and provides diagnostics. It does not guarantee delivery, sequencing, or retransmission, so it is not used for actual data transfer.

---

### Q3: What is the role of convergence in routing stability?

Convergence ensures all routers share a consistent network view, preventing routing loops, black holes, and incorrect forwarding decisions.

---

### Q4: Why is link-state routing faster than distance vector routing?

Because link-state protocols share topology changes immediately and compute shortest paths using SPF algorithm instead of waiting for periodic updates.

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# Section C: Long Questions



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### Q1: Explain how routers determine best path in a dynamic network.

Routers evaluate multiple routes using metrics defined by routing protocols. Each protocol assigns values like hop count, bandwidth, or cost. The router compares all available routes and selects the one with the lowest metric. If multiple equal paths exist, load balancing is used. If no route exists, a default route or ICMP unreachable message is triggered.

---

### Q2: Explain difference between distance vector and link-state routing with reasoning.

Distance vector routing depends on neighbor-based information and periodic updates, making it simpler but slower and prone to loops. Link-state routing builds a full topology map using LSP flooding and calculates shortest path using SPF algorithm, making it more scalable, accurate, and faster in convergence.

---

### Q3: How do ping and traceroute help in troubleshooting networks?

Ping checks basic connectivity using ICMP echo messages and measures response time. Traceroute identifies each hop between source and destination using TTL values, helping locate failure points or delays in the network path.

## Q4: Explain why static routing is still used despite dynamic routing availability.

Static routing is used for security, stability, and efficiency in small or stub networks. It consumes no bandwidth for updates and provides full control. It is also used for default routes, backup paths, and route summarization.